

The Aging Brain

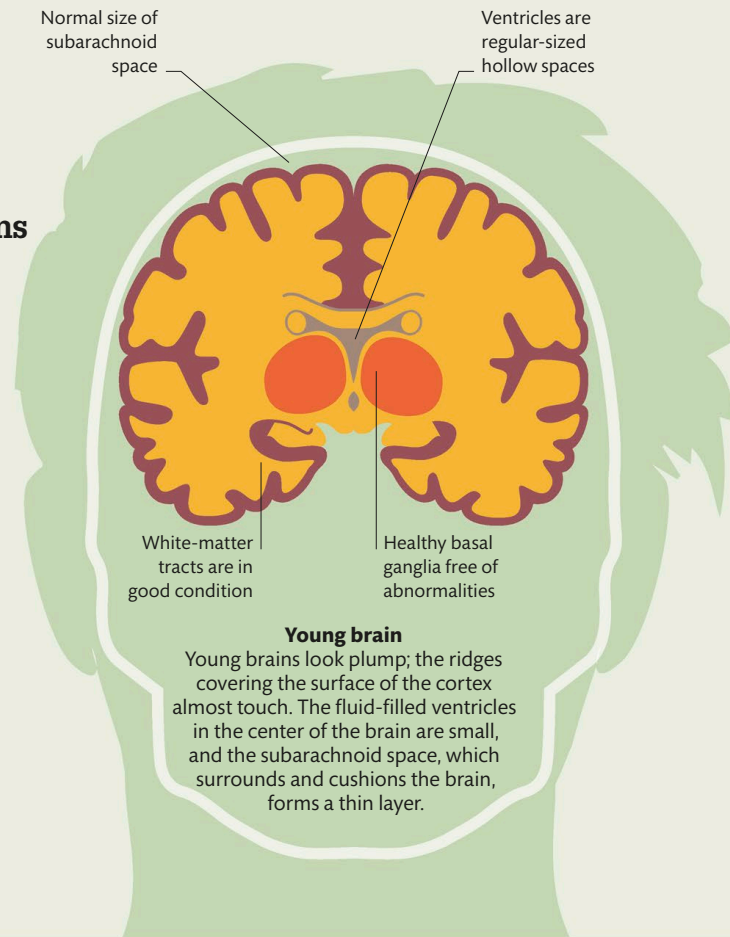
With age, some abilities decline as neurons degenerate and the brain decreases in volume. In those neurons that remain, impulses may travel more slowly.

The shrinking brain

As we age, there is a natural reduction of neurons as they degenerate, and the brain as a whole shrinks 5 to 10 percent in volume. This is partially due to decreased blood flow to aging brains. The fatty myelin that insulates the axons of neurons also decays with age, leaving brain circuits less efficient at transmitting information, which can lead to problems with memory recall and maintaining balance.

KEY

- Gray matter
- Basal ganglia
- White matter
- Ventricles

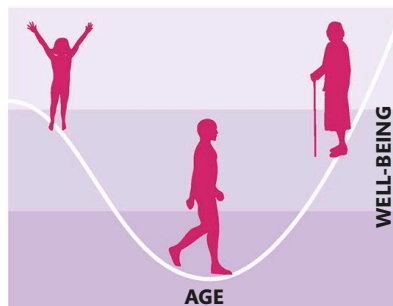


Aging and happiness

Aging might seem like a bad thing, but studies have shown that as we get older, our feelings of happiness and well-being increase, while levels of stress and worry decrease. Older adults' brains seem to be better at focusing on the positive. They are more likely to remember happy than sad pictures and spend more time looking at happy faces than angry or upset ones.

Ups and downs

A study found younger and older people reported higher levels of well-being than those in middle age. Happiness levels rose steadily from age 50 onward.



ALZHEIMER'S DISEASE

Alzheimer's disease, the most common form of dementia (see p.200), is linked to the buildup of proteins in the brain, which clump into plaques and tangles. Eventually, affected brain cells die, causing memory loss and other symptoms. Scientists do not know yet whether the proteins cause the disease or are a symptom of it, and drugs to break them down have not helped patients.

